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Inventor(s)

UZEL; Baptiste C. et al.

REMOTE SURFACE TOUCH DETECTION FOR ELECTRONIC DEVICES

Abstract

Implementations of the subject technology provide systems and methods for detection of a user input with an electronic device, in which the input is received on or using a case for an electronic device. In one or more implementations, the case may be free of user input components such as buttons, switches, or touch sensors.

Inventors: UZEL; Baptiste C. (San Diego, CA), DURFEE; Kevin (San Francisco, CA), WATSON; Robert D. (Menlo Park, CA), PAEK; Timothy S. (Mercer Island, WA), JAMAL; Wasifa (San Diego, CA), BANERJEE; Soham (San Jose, CA), DHULKHED; Vaishnavi (San Francisco, CA)

Applicant: Apple Inc. (Cupertino, CA)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION(S) [0001] The present application claims the benefit of U.S. Provisional Application No. 63/561,151, entitled “REMOTE SURFACE TOUCH DETECTION FOR ELECTRONIC DEVICES”, filed Mar. 4, 2024, the entirety of which is incorporated herein for reference.

TECHNICAL FIELD

[0002] The present description relates generally to user inputs to electronic devices including, for example, to remote surface touch detection for electronic devices.

BACKGROUND

[0003] Audio devices such as headphones and earbuds can include speakers for outputting sound to a user's ears, and microphones for capturing the sound of the user's voice. In some instances, audio devices can be stored in a case.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] Certain features of the subject technology are set forth in the appended claims. However, for purpose of explanation, several embodiments of the subject technology are set forth in the following figures.

[0005] FIG. 1 illustrates an example system architecture including various electronic devices that may implement the subject system in accordance with one or more implementations.

[0006] FIG. 2 illustrates a schematic diagram of a case within which other electronic devices are disposed in accordance with one or more implementations.

[0007] FIG. 3 illustrates a schematic cross-sectional diagram of a case within which media output devices are disposed in accordance with one or more implementations.

[0008] FIG. 4 illustrates a schematic diagram of an example operation for remote surface touch detection by a media output device within a case in accordance with one or more implementations of the subject technology.

[0009] FIG. 5 illustrates a schematic diagram of an example operation for remote surface touch detection by multiple media output devices within a case in accordance with one or more implementations of the subject technology.

[0010] FIG. 6 illustrates a schematic diagram of another example operation for remote surface touch detection by multiple media output devices within a case in accordance with one or more implementations of the subject technology.

[0011] FIG. 7 illustrates a schematic diagram of an example operation for surface touch detection by a case in accordance with one or more implementations of the subject technology.

[0012] FIG. 8 illustrates a flow diagram for an example process for surface touch detection for a case for an electronic device in accordance with implementations of the subject technology.

[0013] FIG. 9 illustrates a flow diagram for an example process for remote surface detection by an electronic device in a case in accordance with implementations of the subject technology.

[0014] FIG. 10 illustrates a flow diagram for an example process for surface touch detection by a case for an electronic device in accordance with implementations of the subject technology.

[0015] FIG. 11 illustrates an electronic system with which one or more implementations of the

subject technology may be implemented.

DETAILED DESCRIPTION

[0016] The detailed description set forth below is intended as a description of various configurations of the subject technology and is not intended to represent the only configurations in which the subject technology can be practiced. The appended drawings are incorporated herein and constitute a part of the detailed description. The detailed description includes specific details for the purpose of providing a thorough understanding of the subject technology. However, the subject technology is not limited to the specific details set forth herein and can be practiced using one or more other implementations. In one or more implementations, structures and components are shown in block diagram form in order to avoid obscuring the concepts of the subject technology.

[0017] Aspects of the disclosure may provide for detection of touch inputs on an outer surface of a case for an electronic device, such as a case for earbuds. This can facilitate providing the case without external input components, such as buttons or switches, while also providing case functionality, such as for initiating pairing and/or factory reset operations for the case and/or the electronic device.

[0018] As described in further detail hereinafter, remote surface touch detection operations can be performed by a case for an electronic device when the electronic devices is in the case, by the electronic device that is within the case, and/or by the case when no electronic device is within the case. As one example, in a scenario in which the case is an earbuds case, if no earbuds are present in the case, the case may perform its own touch detection using an accelerometer and a machine learning model at the case. As another example, when one or both earbuds is present in the case, the case may instruct one or both of the earbuds to perform touch detection for the case, and then indicate to the case when a touch (e.g., any touch or a specific type of touch, such as a double tap) is detected. Aspects of the subject technology can also be applied to detect other inputs to or using a case for an electronic device, such as a shake or other movement of the case itself.

[0019] FIG. 1 illustrates an example system architecture **100** including various electronic devices that may implement the subject system in accordance with one or more implementations. Not all of the depicted components may be used in all implementations, however, and one or more implementations may include additional or different components than those shown in the figure. Variations in the arrangement and type of the components may be made without departing from the spirit or scope of the claims as set forth herein. Additional components, different components, or fewer components may be provided.

[0020] The system architecture **100** includes a media output device **150**, an electronic device **104** (e.g., a handheld electronic device such as a smartphone or a tablet), an electronic device **110**, a case **112** (e.g., a smart case), an electronic device **115**, and a server **120**, some or all of which communicatively coupled together by a network **106** (e.g., a local or wide area network). For explanatory purposes, the system architecture **100** is illustrated in FIG. 1 as including the media output device **150**, the electronic device **104**, the electronic device **110**, the case **112**, the electronic device **115**, and the server **120**; however, the system architecture **100** may include any number of electronic and/or audio devices and any number of servers or a data center including multiple servers.

[0021] The media output device **150** may be implemented as an audio device such as a smart speaker, headphones (e.g., a pair of speakers mounted in speaker housings that are coupled together by a headband), or an earbud (e.g., an earbud of a pair of earbuds each having a speaker disposed in a housing that conforms to a portion of the user's ear) configured to be worn by a user (also referred to as a wearer when the audio device is worn by the user), or may be implemented as any other device capable of outputting audio, video and/or other types of media (e.g., and configured to be worn by a user). Each media output device **150** may include one or more speakers such as speaker **151** configured to project sound into an ear of the user **101**, and one or more microphones such as microphone **152** configured to receive audio input such as external noise input and/or external

voice inputs. The media output device **150** may include a power source, such as a battery and/or a wired or wireless power source.

[0022] In one or more implementations, the media output device **150** may include display components for displaying video or other media to a user, or may be provided without a display components. Although not visible in FIG. **1** (see, e.g., FIG. **2** or FIG. **3**), each media output device **150** may include processing circuitry (e.g., including memory and/or one or more processors, such as a central processing unit and/or one or more digital signal processors (DSPs)) and communications circuitry (e.g., one or more antennas, etc.) for receiving and/or processing audio content from one or more of the electronic device **104**, the electronic device **110**, the electronic device **115**, and/or the server **120**. The processing circuitry of the media output device **150** or another device may operate one or more speakers, such as the speaker **151**, to generate the sound. The memory may store one or more machine learning models implemented as neural networks and trained for one or more audio processing operations and/or one or more detection operations for detecting user inputs to the media output device **150** and/or to a remote surface, such as a surface of the case **112**.

[0023] For example, case **112** may be a case for the media output device **150**. For example, the case **112** may be configured to temporarily and/or occasionally store the media output device **150**, such as when the media output device **150** is not in use (e.g., not being used to generate sound with one or more speakers thereof, and/or not being worn by the user **101**). For example, the case **112** may provide mechanical and/or environmental protection for the media output device **150** when the media output device **150** is stored within the case **112**. In one or more implementations, the case **112** may be configured to charge a battery of the media output device **150** when the media output device **150** is stored in the case **112**. The case **112** may be referred to herein as an enclosure, a box, a carrier, a charging carrier, a jacket, a holder or the like.

[0024] The media output device **150** may include communications circuitry for communications (e.g., directly or via network **106**) with the electronic device **104**, the electronic device **110**, the case **112**, the electronic device **115**, and/or the server **120**, the communications circuitry including, for example, one or more wireless interfaces, such as WLAN radios, cellular radios, Bluetooth radios, Zigbee radios, near field communication (NFC) radios, and/or other wireless radios. The electronic device **104**, the electronic device **110**, the case **112**, the electronic device **115**, and/or the server **120** may include communications circuitry for communications (e.g., directly or via network **106**) with media output device **150** and/or with the others of the electronic device **104**, the electronic device **110**, the case **112**, the electronic device **115**, and/or the server **120**, the communications circuitry including, for example, one or more wireless interfaces, such as WLAN radios, cellular radios, Bluetooth radios, Zigbee radios, near field communication (NFC) radios, and/or other wireless radios. In one or more implementations, the case **112** may have a limited set of communications circuitry as compared with the communications circuitry of the electronic device **104**, the electronic device **110**, the electronic device **115**, or the media output device **150**. For example, the case **112** may include only Bluetooth circuitry, or only Bluetooth and WiFi circuitry in some implementations.

[0025] The media output device **150** may be communicatively coupled to a companion device such as the electronic device **104**, the electronic device **110** and/or the electronic device **115** in some use cases. Media output device **150** may operate in various modes. For instance, the media output device **150** can operate in various modes of operations, such as a transparent mode of operation in which at least portions of an external audio input to the media output device **150** is passed through to the ear of the user using the speaker(s) **151**, or a noise-cancelling mode of operation in which the media output device **150** removes or cancels all external audio input (e.g., by filtering out external audio input and/or by generating an out-of-phase noise cancelling signal to cancel out the audio input), with or without concurrently outputting audio content in either of these modes of operation. In the transparent mode operation and/or other modes of operation such as a voice enhancement

mode of operation, one or more DSPs and/or neural networks of the media output device may perform source separation operations on incoming external audio input and may remove, cancel, suppress, and/or enhance various components of the separated incoming external audio input. In the noise-cancelling mode of operation, one or more DSPs and/or neural networks of the media output device may perform source separation operations on the incoming external audio input to suppress, cancel, or remove all of the incoming external audio input from the sound that enters the user's ear.

[0026] The media output device **150** may also operate in one or more other modes of operation, such as a call/conference mode in which one or more DSPs and/or neural networks separate the voice of the user of the media output device **150** from other sounds in an audio input for transmission to another device (e.g., a remote device participating in a call, an audio conference, and/or a video conference with the user **101**), or a speaker enhancement or hearing aid mode of operation in which one or more DSPs and/or neural networks separate the voice of a speaker other than the user of the media output device (or another predetermined sound such as an alarm, a cry of a baby or a pet, etc.) from other sounds in the audio input and the speaker(s) **151** of the media output device **150** are used to output the voice of the speaker (or the other predetermined sound) to the car(s) of the user **101**.

[0027] Media output device **150** may also include one or more sensors such as touch sensors, inertial sensors (e.g., accelerometers, gyroscopes, and/or magnetometers), environmental sensors, and/or force sensors, such as for receiving user input. For example, a user/wearer of media output device **150** may tap a touch sensor of the media output device **150** or pinch a force sensor of the media output device **150** briefly to control the audio content being played, to control volume of the playback, and/or to switch between modes of operation. In one or more implementations, the user may hold down the force sensor while the media output device is operated in the noise-cancelling mode of operation to temporarily switch to the transparent mode of operation until the force sensor is released.

[0028] The electronic device **104** may be, for example, a smartphone, a portable computing device such as a laptop computer, a peripheral device (e.g., a digital camera, headphones, another audio device, or another media output device), a tablet device, a wearable device such as a smart watch, a smart band, and the like, any other appropriate device that includes, for example, processing circuitry and/or communications circuitry for pairing with, connecting to, and/or providing audio content to media output device(s) **150**. In FIG. 1, by way of example, the electronic device **104** is depicted as a mobile smartphone device with a touchscreen. In one or more implementations, the electronic device **104** and/or the media output device **150** may be, and/or may include all or part of, the electronic device discussed below with respect to the electronic system discussed below with respect to FIG. 11.

[0029] The electronic device **115** may be, for example, desktop computer, a portable computing device such as a laptop computer, a smartphone, a peripheral device (e.g., a digital camera, headphones, another audio device, or another media output device), a tablet device, a wearable device such as a watch, a band, and the like. In FIG. 1, by way of example, the electronic device **115** is depicted as a desktop computer. The electronic device **115** may be, and/or may include all or part of, the electronic system discussed below with respect to FIG. 11.

[0030] The server **120** may form all or part of a network of computers or a group of servers **130**, such as in a cloud computing or data center implementation. For example, the server **120** stores data and software, and includes specific hardware (e.g., processors, graphics processors and other specialized or custom processors) for rendering and generating content such as graphics, images, video, audio and multi-media files for computer-generated reality environments. In an implementation, the server **120** may function as a cloud storage server.

[0031] FIG. 2 illustrates a schematic diagram of the case **112**, in a use case in which the case **112** stores two electronic devices **204**. As examples, the electronic devices **204** of FIG. 2, may be

implementations of the media output device **150**, the electronic device **104**, the electronic device **115**, or any other electronic device that can be stored in a case. For example, the case **112** may include a housing **200** within which one or more cavities **270** are defined. For example, the housing **200** may be sufficiently small to be carried in the hand, a pocket, or a bag of a user (e.g., the user **101**). In one or more implementations the housing **200** may include a lid **212** or cap that can be removed and/or otherwise opened (via a hinge) to provide access to the cavities **270**, and can be closed to enclose the electronic devices **204** in the case **112**. Cavities **270** may each be configured to receive one of the electronic devices **204**. For example, the cavities **270** may each have a size and/or a shape that conforms to a size and/or shape of a housing **205** of the electronic device **204** that the cavity **270** is designed to receive. For example, an outer surface **279** of the housing **205** of each of the electronic devices **204** may contact an interior surface of the cavity **270** in the case **112** at one or more locations, when the electronic devices **204** are within the cavities **270**.

[0032] As shown, the case **112** itself may be an electronic device (e.g., a smart case) that includes processing circuitry (e.g., one or more processors, such as processor **224**), memory **226**, communications circuitry **222**, one or more sensors **220**, and/or a power source such as battery **228**. In one or more implementations, the case **112** may be configured to charge a battery **208** of each of the electronic devices **204** when the electronic devices **204** are disposed within the cavities **270**. For example, in one or more implementations, the electronic devices **204** may have external electrical contacts that each contact a corresponding electrical contact on the interior surface of the cavity **270** when the electronic devices **204** are disposed within the cavities **270**. Electrical charge, current, voltage, and/or communication signals may be provided between the electronic devices **204** and the case **112** via the electrical contacts when the electronic devices **204** are disposed within the cavities **270**.

[0033] As shown, in one or more implementations, the case **112** may be free of (e.g., provided without, or exclude) any user input components, such as buttons, switches, or touch sensors (e.g., capacitive or resistive touch sensors), including on the housing **200** (e.g., on the outer surface **299** of the housing **200**). For example, the sensor(s) **220** may be free of (e.g., may not include or may exclude) buttons, switches, touch sensors, or any other sensors or components on the outer surface **299** of the housing **200**. The sensors **220** may include one or more inertial sensors, such as one or more accelerometers, one or more gyroscopes, and/or one or more magnetometers. For example, the sensor(s) **220** may be configured for detecting the orientation and/or motion of the case **112**. However, it may be desirable to be able to receive user inputs to the case **112**, even when the case **112** is provided without any user input components on the outer surface **299** of the housing **200**.

[0034] In one or more implementations, the case **112** may detect a user input to the outer surface **299** of the housing **200** using the sensor(s) **201**, such as by providing an accelerometer signal from an accelerometer of the sensor(s) **201** to a machine learning model running on the processor **202**, the machine learning model having been trained to detect touch inputs (e.g., taps, double taps, triple taps, other taps, swipes, or other touch inputs) to the case **112** based on accelerometer data. For example, when a user taps the outer surface **299** of the housing **200** (e.g., at any location on the housing **200**), the case **112** may move, and/or vibrate in a characteristic manner that can be represented in the accelerometer data and that can be learned by the machine learning model at the case **112**. However, because the case **112** is primarily a storage container for the electronic devices **204** (e.g., rather than a standalone electronic device configured to generate user outputs and/or run applications for a user), the processing power of the processor **224** may be limited. Accordingly, it may be beneficial to be able to utilize one or both of the electronic devices **204** to detect touch inputs (e.g., taps, double taps, triple taps, other taps, swipes, or other touch inputs) to the case **112** (e.g., touch inputs on a remote surface that is remote from the outer surface **299** of the electronic device(s) **204** themselves).

[0035] As shown, each of the electronic devices **204** may include processing circuitry, such as one or more processors **202**, memory **244**, a power source such as a battery **208**, and communications

circuitry **206**. For example, the communications circuitry **206** of each electronic device **204** may be configured for communicating with the case **112**, the other of the electronic devices **204**, the electronic device **104**, the electronic device **110**, the electronic device **115**, and/or other electronic devices within a wireless communications range of the electronic device **204**.

[0036] As shown, each of the electronic devices **204** may also include one or more sensor(s) **201** (e.g., touch sensors, force sensors, pressure sensors, environmental sensors, and/or inertial sensors such as accelerometers, gyroscopes, and/or magnetometers). As shown, the electronic devices **204** may each also include one or more speakers **151** (e.g., for generating sound for output for a user), and/or one or more microphones **152** for receiving audio input signals. In the example of FIG. 2, each of the electronic devices **204** includes two microphones **152** (e.g., a top microphone **152** and a bottom microphone **152B** in some examples). In one or more implementations, one or more machine learning models may be stored by the memory **244**. In one or more implementations, the machine learning models at each of the electronic devices **204** may have been trained to detect one or more user inputs (e.g., touch inputs) to the outer surface **299** of the housing **200** of the case **112** when the electronic device(s) **204** are within the case **112** (e.g., stored in the cavities **270** within the housing **200**), even though the user inputs are not received directly at either of the electronic devices **204**.

[0037] For example, in one or more implementations, one or both of the electronic devices **204** may detect a user input to the outer surface **299** of the housing **200** of the case **112** using the sensor(s) **201** and/or the microphones **152**, such as by providing an accelerometer signal from an accelerometer of the sensor(s) **201** and/or one or more microphone signals from the microphones **152** to a machine learning model running on the processor **202**, the machine learning model having been trained to detect user inputs (e.g., touch inputs, such as taps, double taps, triple taps, other taps, swipes, or other touch inputs, and/or other user inputs such as shakes of the case **112**) on a remote surface (e.g., user inputs to the housing **200** of the case **112**) based on accelerometer data and/or microphone data. For example, when a user taps the outer surface **299** of the housing **200** (e.g., at any location on the housing **200**), the electronic devices **204** may move, and/or vibrate, within the case **112**, in a characteristic manner than can be represented in the accelerometer data, and that can be learned by the one or more machine learning models at the electronic device(s) **204**. When the user taps the outer surface **299** of the housing **200** (e.g., at any location on the housing **200**), the sound of the user tap(s) and/or the movement of the electronic devices **204** within the case **112** may generate sounds that can be represented in the microphone data, and that can be learned by the one or more machine learning models at the electronic device(s) **204**.

[0038] As shown in FIG. 2, the case **112** may also include one or more output components, such as output component **251**, in one or more implementations. For example, the output component **251** may include a visible output component (e.g., one or more light-emitting diodes (LEDs)), and/or one or more audio output components (e.g., a speaker) configured to generate outputs that provide indications to a user of received user input (e.g., to provide visual feedback by turning on and/or flashing an LED or to provide audio feedback by providing a ding, ping, or chime from a speaker). For example, the output component **251** may include a simple speaker that has a reduced range of output (e.g., a single tone, or two or several tones), relative to the speaker(s) **151** of the electronic devices **204**, which may be able to output music, podcasts, or any other audio content. In one or more implementations, the media output device(s) **150** in the case **112** may also, or alternatively, generate outputs that provide indications to a user of received user input to the case **112**. For example, the speaker(s) of the media output device **150** may be used to generate an audible indicator of the received user input and/or to vibrate the media output device(s) **150** to provide an auditory and/or haptic indication of the received user input to the case **112**. In one or more implementations, the media output device(s) **150** may include haptic components for generating haptic feedback to user inputs to the case **112**.

[0039] As discussed herein, the cavities **270** may each have a size and/or a shape that conforms to a

size and/or shape of a housing **205** of the electronic device **204** that the cavity **270** is designed to receive. For example, as shown in FIG. **3**, the case may be a case for a pair of earbuds (e.g., a pair of media output devices **150** implemented as earbuds), and the cavities **270** each may have a size and shape that conforms to the size and shape of an earbud. In this example, the size and the shape of the housings **205** of each of the media output devices **150** may be configured to be seated within the concha of a user's ear and to project sound directly into the ear canal of the user's ear.

[0040] FIGS. **4-7** illustrate various use cases in which user inputs to the housing **200** of the case **112** are detected. In the examples of FIGS. **4-7**, the case **112** is described as a case for media output devices **150** that are implemented as earbuds, and the user inputs are described as taps or other touches on the outer surface **299** of the housing **200** of the case **112**. However, it is appreciated that the case **112** may be implemented as a case for any other electronic device, and the user input may be other user inputs to the case, such as swipes across the outer surface **299** of the case **112**, or shakes or other movements of the overall case **112** itself.

[0041] For example, FIG. **4** illustrates a use case in which the case **112** is implemented as a case for media output devices **150** that are implemented as earbuds, and in which one media output device **150** is in one of the cavities **270** in the case **112** and in communication with the case **112** (e.g., via the communications circuitry **222** of the case **112**). In this example, the other cavity **270** in the case **112** is empty. However, this is merely illustrative, and the operations of FIG. **4** can be performed while a second media output device **150** is in the other of the cavities **270** in some scenarios.

[0042] As illustrated in FIG. **4**, the case **112** (e.g., the processor **224**) may determine (**400**) that the media output device **150** is in the case **112** (e.g., in the cavity **270**). In one or more implementations, the case **112** may determine that the media output device **150** is in the case **112** responsive to a detection **401** (e.g., by the processor **224**) that the lid **212** has been opened. For example, the processor **224** may initiate an operation to determine whether one or both of the media output devices **150** are in the case **112** each time the processor **224** detects that the lid **212** has been opened. Determining that the media output device **150** is in the case **112** may include determining whether an electrical contact of the media output device **150** is in contact with a corresponding electrical contact within the cavity **270** (e.g., by determining whether the processor **224** can send and/or receive communications from the processor **202** via the electrical contacts). In one or more other implementations, the processor **224** may determine whether the media output device **150** is in the case **112** based on a wireless communication (e.g., a near field communications (NFC) communication, a Bluetooth communication, and/or a WiFi communication) with the media output device **150**.

[0043] As shown, once the case **112** has determined that the media output device **150** is in the case, the case **112** may provide (e.g., using the communications circuitry **222**, as controlled by the processor **224**) an instruction **402** to the media output device **150** to perform one or more detection operations for detecting a user input to the case **112** (e.g., to the outer surface **299** of the housing **200** of the case **112**). As shown, a user (e.g., user **101**) may use their finger **404** or another instrument to provide a user input, such as one or more taps, touches, or swipes, on the outer surface **299** of the housing **200**. As shown, responsive to a detection, by the media output device **150**, of the user input (e.g., by the finger **404** or other instrument of a user on the outer surface **299** of the housing **200**) to the case **112**, the media output device **150** may provide an indication **406**, to the case **112**, of the detected user input.

[0044] For example, responsive to the instruction **402** (or the detection **401**), the media output device **150** may activate one or more sensors **201** and/or one or more microphones **152** of the media output device **150**. As shown, the sensor(s) **201** may provide sensor data **495** to a machine learning model **499** (e.g., a machine learning model being executed by the processor **202** of the media output device **150**), and/or the microphone(s) **152** may provide audio data **497** to the machine learning model **499**. In one or more implementations, the sensor(s) **220** of the case **112** (e.g., an accelerometer) may also provide (e.g., via the communications circuitry **222**) sensor data

to the machine learning model **499** at the media output device **150** for use in detecting the user input. Based on an output of the machine learning model **499**, the media output device **150** may provide the indication **406** of the user input to the case **112**.

[0045] As shown, responsive to receiving the indication **406**, the case **112** may provide an instruction **409** to the media output device **150**. For example, the instruction **409** may be an instruction to provide a battery status, or to initiate a pairing operation. In one or more implementations, the instruction **409** may be based on the user input detected (e.g., a single tap, a double tap, a triple tap, a swipe, a shake, etc.) by the machine learning model **499**. In one or more implementations, the media output device **150** may initiate the pairing operation, such as by transmitting (e.g., with the communications circuitry **206**) a signal **408** (e.g., a Bluetooth signal or other wireless signal) that can be detected by a nearby electronic device (e.g., electronic device **104**) for pairing with the media output device **150**.

[0046] In one or more implementations, the instruction **409** may be an instruction to initiate the pairing operation after a period of time (e.g., one, two, or several seconds, such as to allow time for an additional user input to the case that, combined with the first user input, changes the intention and the result of the user input). For example, as shown in FIG. **4**, the media output device **150** may, after providing the indication **406** of the user input to the case **112** and receiving the instruction **409**, identify (e.g., using additional sensor data **495**, additional audio data **497**, and/or additional sensor data from the sensor **220** provided to the machine learning model **499**) another user input to the case **112**, and provide, to the case **112**, another indication **410** of the other user input to the case **112**. As shown, responsive to the indication **410**, the case **112** may provide, to the media output device **150**, an instruction **412**. For example, the instruction **412** may be an instruction to perform an action responsive to the detection of the other user input indicated by the indication **410** and/or the first user input indicated by the indication **406**.

[0047] As illustrated in FIG. **4**, responsive to the instruction **412**, the media output device **150** may perform an action **414**. In the example of FIG. **4**, the action **414** is a self-contained action that is performed locally (e.g., by the processor **202**) at the media output device **150**, such as a factory reset operation. However, in other examples, the action **414** may be a communicative action that includes transmitting and/or receiving information to and/or from one or more other devices, including another of the media output devices **150** (e.g., within or outside the case **112**), and/or another electronic device, such as the electronic device **104**, the electronic device **110**, the electronic device **115**, and/or the server(s) **120**. As shown, in one or more implementations, responsive to the indication **410**, the case **112** (e.g., the processor **224**) may also perform one or more actions **416** at the case, such as a factory reset of the case. In one or more implementations, the output component **251** may be operated (e.g., turned on, flashed, etc.), during any or all of the operations of FIG. **4**, to indicate the status of the user input and/or to indicate a time-limited opportunity to provide additional user input to the case **112**.

[0048] In the example of FIG. **4**, the detection operations for detecting the user input(s) to the case **112** are performed by one media output device **150** in one cavity **270** within the case **112**. In one or more implementations, when two media output devices **150** are present within the case **112**, the case **112** may determine (e.g., based on the available sensors and/or microphones, and/or based on a charge levels of the batteries in the media output devices) which of the two media output devices **150** to instruct to perform the detection operations of FIG. **4**. However, in other scenarios, both of two media output devices **150** in two cavities **270** within the case **112** may perform detection operations for detecting user input(s) to the case **112**. In this way, the additional sensors, microphones, and/or processing power of the second media output device **150** can be used together with those of the first media output device **150** to enhance the sensitivity and/or accuracy of the user input detections.

[0049] In various implementations, in order to utilize two media output devices for user input detection, the case **112** may instruct one of the media output devices to perform the detection

operations and the one of the media output devices can control the other of the media output devices during the detection operations (e.g., as depicted in the example of FIG. 5), or the case **112** may instruct both of the media output devices to perform the detection operations in parallel (e.g., as depicted in the example of FIG. 6).

[0050] For example, as shown in FIG. 5, responsive to the instruction **402**, the media output device **150** that receives the instruction **402** may exchange communications **500** with the other media output device **150** in the case **112** to perform detection operations for detecting a user input (e.g., with the finger **404** or other instrument of the user) to the housing **200** of the case **112**. In one or more implementations, responsive to the communications **500**, the other media output device **150** may also execute its own machine learning model **499** using sensor data **595** from its own sensors **201** and/or audio data **597** (e.g., microphone data) from its own microphones **152** (e.g., and/or sensor data from the sensor **220** of the case **112**), and provide an output of its machine learning model **499** to the media output device **150** that received the instruction **402**. The media output device **150** that received the instruction **402** may then combine the output received from the other media output device **150** with the output of its own machine learning model **499** to determine whether to send the indication **406** to the case. In one or more other implementations, responsive to the communications **500**, the other media output device **150** may also, or alternatively, provide sensor data **595** from its sensors **201** and/or audio data **597** from its microphones **152** to the machine learning model **499** at the media output device **150** that received the instruction **402**. In these other implementations, the machine learning model **499** at the media output device **150** that received the instruction **402** may receive the sensor data **595** and the audio data **597** as inputs along with the sensor data **495** and/or the audio data **497** generated at the media output device **150** that received the instruction **402**, and generate a user input detection and/or classification output based thereon.

[0051] In the example of FIG. 5, the instruction **409** may be provided to the media output device that received the instruction **402**. The media output device **150** that received the instruction **402** may exchange communications **502** with the other media output device **150** in the case **112** to instruct the other media output device **150** in the case **112** to perform the pairing operations (e.g., using pair information received by the media output device **150** that received the instruction **402**, or by transmitting its own pairing signal **503** that can be detected by a nearby electronic device). As shown, communications **504** (e.g., including an output of a machine learning model at the other media output device **150**, or sensor data and/or microphone data from the other media output device **150**) may be exchanged between the media output device **150** that received the instruction **402** and the other media output device **150** in the case **112**, to perform additional detection operations to detect an additional user input at the case **112** (e.g., within a predetermined period of time after the detection of the first user input to the case **112**).

[0052] Detection, by the media output devices **150**, or an additional user input may result in the indication **410** being provided to the case **112** to indicate the detection of the additional user input. As in FIG. 4, responsive to receiving the indication **410**, the case **112** may provide the instruction **412** to the media output device **150** that received the instruction **402**. In the example of FIG. 5, the media output device **150** that received the instruction **402** may exchange communications **506** with the other media output device **150** in the case **112** to instruct the other media output device **150** to also perform the action **414**.

[0053] In the example of FIG. 5, the second media output device **150** in the case **112** performs detection operations under control of the first media output device **150** that communicates with the case **112**. FIG. 6 illustrates another example in which the case **112** may instruct both of the media output devices **150** that are in the case **112** to perform the detection operations in parallel. For example, as shown in FIG. 6, in one or more implementations, the instruction **402** (e.g., to perform the one or more detection operations) may be provided by the case **112** to each of the (e.g., two) media output devices **150** in the case **112**).

[0054] As indicated, each of the two media output devices **150** in the case **112** may operate their own machine learning model **499** (e.g., in parallel with the operation of the machine learning model **499** at the other of the media output devices **150**). For example, the machine learning model **499** at a first one of the media output device **150** may receive sensor data **495** from the sensor(s) **201** and/or audio data **497** and/or microphone(s) **152** at that media output device **150**, and the machine learning model **499** at a second one of the media output device **150** may receive sensor data **595** from the sensor(s) **201** and/or audio data **597** and/or microphone(s) **152** at that media output device **150**. Although not depicted in FIG. 6, the machine learning model **499** at one or both of the media output devices **150** may also receive sensor data from one or more sensors **220** of the case **112**.

[0055] As shown, the indication **406** of the detected user input may be provided to the case **112** from one of the media output devices **150** in the case **112**, and an indication **606** of the detected user input may also be provided to the case **112** from the other of the media output devices **150** in the case **112**. In this example, the case **112** (e.g., the processor **224**) may use the indication **406** and the indication **606** (e.g., alone or in combination) to determine whether to provide the instruction **409** to the media output devices **150** (e.g., to initiate a pairing operation). For example, as in the example of FIG. 5, responsive to the instruction **409** from the case **112**, a first one of the media output devices **150** may initiate the pairing operation, such as by transmitting (e.g., with the communications circuitry **206** of the first media output device) a signal **408** that can be detected by a nearby electronic device for pairing with the first media output device **150**, and a second one of the media output devices **150** may initiate the pairing operation, such as by transmitting (e.g., with the communications circuitry **206** of the second media output device) a signal **503** that can be detected by a nearby electronic device for pairing with the second media output device **150**.

[0056] As shown in FIG. 6, each of the (e.g., two) media output devices **150** in the case **112** may perform additional detection operations to detect an additional user input at the case **112** (e.g., within a predetermined period of time after the detection of the first user input to the case **112**), which may result in the indication **410** being provided to the case **112** from a first one of the media output devices and/or an indication **610** being provided to the case **112** from a second one of the media output devices **150** to the case **112** to indicate the detection by one or both of the media output devices **150** of the additional user input. Responsive to receiving the indication **410** and/or the indication **610**, the case **112** may provide the instruction **412** to both of the media output devices **150** in the case **112**. The media output devices **150** may each perform the action **414** responsive to the instruction **412**.

[0057] In each of the examples of FIGS. 4, 5, and 6, at least one media output device **150** is present in the case **112**. However, in one or more use cases, the case **112** may perform its own user input detections when no media output devices **150** are present in the case.

[0058] For example, as illustrated in FIG. 7, the case **112** (e.g., the processor **224**) may determine (**700**) (e.g., responsive to a detection **401** that the lid **212** of the case **112** has been opened), that no media output devices **150** are present in the case **112** (e.g., in the cavities **270**). As shown, responsive to the determination (**700**) that no media output devices **150** are present in the case **112**, the case **112** (e.g., the processor **224**) may obtain sensor data **702** from one or more sensors **220** (e.g., accelerometers, gyroscopes, and/or magnetometers) of the case **112**.

[0059] For example, the sensor data **702** may be provided as an input to a machine learning model **704** at the case **112** (e.g., a machine learning model being executed by the processor **224**). As indicated in FIG. 7, the case **112** may identify (**703**), based on an output from the machine learning model **704**, a user input on an exterior surface (e.g., outer surface **299**) of the case **112**. In one or more implementations, the case **112** (e.g., the processor **224**) may perform one or more predetermined actions **710** responsive to the identifying of the user input on the exterior surface of the case **112**.

[0060] As shown, in one or more implementations, following a first detection of a user input on the exterior surface of the case **112**, the case **112** may perform additional detection operations for a

period of time after the first detection, in order to detect an additional user input on the exterior surface of the case **112**. In one or more implementations, the case **112** may perform the one or more predetermined actions **710** responsive to the first and the second detected user inputs on the exterior of the case **112** (e.g., the second user input may be a confirmatory input for the predetermined action), or the case **112** may perform a different predetermined action if the second user input is detected on the exterior surface of the case **112** within the predetermined period of time.

[0061] As examples, the predetermined action **710** may include performing a factory reset operation for the case **112** or performing a locating operation, such as for locating one or both of the media output devices **150** that are not in the case **112**. For example, the locating operation may include transmitting a signal **712** (e.g., a wireless signal) to the media output device(s) **150** and/or receiving a signal **714** (e.g., a wireless signal) from the media output device(s) **150**, and determining a location of the electronic device relative to the case based on the signal **712** and/or the signal **714** (e.g., based on a strength of the signal **714**, and/or based on location information included in the signal **714**). For example, the signal **712** may be a request for the media output device(s) **150** to respond with a ranging signal and/or a signal containing location information that indicates the (e.g., relative or absolute) location of the media output device(s) **150**. In one or more implementations, the action **710** may include providing an output (e.g., with the output component **251** and/or a speaker of the case) to guide a user of the case to the location of the media output device(s) **150**.

[0062] FIG. **8** illustrates a flow diagram of an example process **800** for operating a case for an electronic device, in accordance with implementations of the subject technology. For explanatory purposes, the process **800** is primarily described herein with reference to the media output device **150** and case **112** of FIGS. **1-7**. However, the process **800** is not limited to the media output device **150** and case **112** of FIGS. **1-7**, and one or more blocks (or operations) of the process **800** may be performed by one or more other components of other suitable devices, including the electronic device **110**, the electronic device **115**, and/or the servers **120**. Further for explanatory purposes, some of the blocks of the process **800** are described herein as occurring in serial, or linearly. However, multiple blocks of the process **800** may occur in parallel. In addition, the blocks of the process **800** need not be performed in the order shown and/or one or more blocks of the process **800** need not be performed and/or can be replaced by other operations.

[0063] As illustrated in FIG. **8**, at block **802**, processing circuitry (e.g., processor **224**) of a case (e.g., case **112**) having an internal cavity (e.g., cavity **270**) for storing for an electronic device (e.g., an electronic device **204**, such as a media output device **150**) may determine that the electronic device is present in the case (e.g., within the internal cavity in the case). An outer surface (e.g., outer surface **299**) of the case may be free of user input components (e.g., the case may not have any buttons, switches, touch sensors, or the like disposed on the outer surface thereof). In one or more implementations, the electronic device may be or include an audio output device (e.g., an earbud or a headphone, such as a wireless earbud or headphone), and the case may be configured to charge a battery (e.g., battery **208**) of the audio output device when the audio output device is in the case. For example, the case may determine that the electronic device is present in the case responsive to a detection (e.g., detection **401**), by the case, of an opening of a lid (e.g., lid **212**) of the case.

[0064] At block **804**, the case may provide a first instruction (e.g., instruction **402**) to the electronic device (e.g., over a wired or wireless connection between the case and the electronic device) to perform one or more detection operations for detecting a user input to the case. For example, the user input to the case may be a touch input (e.g., a touch, tap, double tap, swipe, or other touch input) on the outer surface of the case. For example, the touch input may include one or more taps on the outer surface of the case. In other examples, the user input may be a shake or other movement of the case.

[0065] At block **806**, the case may receive, from the electronic device, an indication (e.g., indication **406**) of the user input to the case. For example, the electronic device that is disposed within the case may detect the user input based on one or more sensor signals (e.g., accelerometer signals, gyroscope signals, and/or other inertial sensor signals, such as sensor signal **495** and/or sensor signal **595**) from one or more sensors (e.g., sensors **220**, such as an accelerometer, a gyroscope, and/or other inertial sensor), and/or based on one or more microphone signals (e.g., audio signals **497** and/or audio signal **597**) from one or more microphones (e.g., microphones **152**, **152B**) of the electronic device. In one or more implementations, the one or more sensor signals and/or the one or more microphone signals may be provided to a machine learning model (e.g., machine learning model **499**) at the electronic device, the machine learning model having been trained to detect user inputs to an outer surface of the case based on one or more microphone signals and/or one or more sensor signals generated by microphones and/or sensors that are housed within a housing (e.g., housing **205**) that is removable from the case.

[0066] In one or more implementations, a case that is free of buttons, switches, touch sensors, or the like disposed on the outer surface thereof may include, on the outer surface, one or more features, such as a contrastive texture or material, that may provide a passive haptic feedback that may indicate a particular location on the case at which a user can tap the case to provide an input, as described herein. For example, the one or more features may be provided at a location that has been determined to enhance the effect, on the sensors and/or microphones of the electronic device (e.g., the media output device **150**), of a tap on the case **112**.

[0067] At block **808**, the case may provide a second instruction (e.g., instruction **409** or instruction **412**) to the electronic device to perform an action (e.g., a pairing operation or other action) responsive to the detection of the user input. For example, the second instruction may include an instruction, to the electronic device from the case, to start a pairing operation after a predetermined amount of time. In one or more implementations, the case may also receive, from the electronic device, an additional indication (e.g., indication **410**) of an additional user input to the case prior to passage of the predetermined amount of time, and may instruct (e.g., instruction **412**) the electronic device to perform an updated action (e.g., action **414**), different from the action (e.g., immediately or if a further additional user input, such as third user input) to the case is detected by the electronic device within an updated predetermined amount of time). For example, the updated action may include a factory reset of at least one of the case or the electronic device.

[0068] In one or more implementations, the case may include an accelerometer (e.g., sensor **220**), and the process **800** may also include providing, by the case, accelerometer data from the accelerometer to the electronic device for use in the one or more detection operations (e.g., as discussed herein in connection with FIG. **4**). For example, the indication (e.g., indication **406** or the indication **410**) of the user input to the case may be based on the accelerometer data and additional accelerometer data obtained by an additional accelerometer (e.g., sensor **201**) of the electronic device.

[0069] In one or more implementations, the case (e.g., processor **224**) may determine that the electronic device and an other electronic device (e.g., another electronic device **204**, such as another media output device **150**) are present in the case, and may determine, prior to providing the first instruction to the electronic device, which of the electronic device and the other electronic device is to perform the one or more detection operations. In one or more implementations, the case may also provide, to the other electronic device, the first instruction (e.g., instruction **402**) to perform the one or more detection operations in parallel with the one or more detection operations at the electronic device (e.g., as discussed herein in connection with FIG. **6**).

[0070] FIG. **9** illustrates a flow diagram of an example process **900** for user input detection using an electronic device that is disposed within a case, in accordance with implementations of the subject technology. For explanatory purposes, the process **900** is primarily described herein with reference to the media output device **150** and the case **112** of FIGS. **1-7**. However, the process **900**

is not limited to the media output device **150** and case **112** of FIGS. **1-7**, and one or more blocks (or operations) of the process **900** may be performed by one or more other components of other suitable devices, including the electronic device **110**, the electronic device **115**, and/or the servers **120**. Further for explanatory purposes, some of the blocks of the process **900** are described herein as occurring in serial, or linearly. However, multiple blocks of the process **900** may occur in parallel. In addition, the blocks of the process **900** need not be performed in the order shown and/or one or more blocks of the process **900** need not be performed and/or can be replaced by other operations.

[0071] As illustrated in FIG. **9**, at block **902**, sensor data (e.g., sensor data **495** and/or audio data **497**) may be obtained from one or more sensors (e.g., sensor(s) **201** and/or microphone(s) **152**) of an electronic device (e.g., electronic device **204**, such as a media output device **150**, such as an earbud) that is disposed within, and removable from, another electronic device (e.g., case **112** having processor **224** and communications circuitry **222**). For example, obtaining the sensor data from the one or more sensors comprises obtaining the sensor data from at least an accelerometer (e.g., sensor **201**) and a microphone (e.g., microphone **152**).

[0072] At block **904**, processing circuitry (e.g., processor **202**) of the electronic device may determine whether the sensor data indicates a user input on an outer surface (e.g., outer surface **299**) of the other electronic device. For example, determining whether the sensor data indicates the user input on the outer surface of the other electronic device may include providing the sensor data to a machine learning model (e.g., machine learning model **499**) at the electronic device, and determining whether the sensor data indicates the user input on the outer surface of the other electronic device based on an output of the machine learning model. For example, the output of the machine learning model may be a binary signal indicating detection or no detection of the user input, or may be a more complex output, such as a classification of one or more user inputs (e.g., as taps, multi-taps, touches, shakes, swipes, etc.) and/or a confidence or probability of the detection of the one or more user inputs. In one or more implementations, obtaining the sensor data may include obtaining the sensor data responsive to an instruction (e.g., instruction **402**) from the other electronic device.

[0073] In one or more implementations, the process **900** may also include providing, from the electronic device to the other electronic device within which the electronic device is disposed, an indication (e.g., indication **406**) of the user input. In one or more implementations, the process **900** may also include receiving, by the electronic device from an additional electronic device (e.g., an additional electronic device **204**, such as an additional media output device **150**, such as a second earbud of a pair of earbuds) disposed within and removable from the other electronic device, additional sensor data (e.g., additional microphone data and/or additional accelerometer data from one or more microphones and/or accelerometers of the other electronic device), and determining, by the electronic device, whether the sensor data indicates the user input on the outer surface of the other electronic device based on the sensor data and the additional sensor data (e.g., by providing the additional sensor data to the machine learning model at the electronic device along with the sensor data obtained by the sensors and/or microphones of the electronic device).

[0074] FIG. **10** illustrates a flow diagram of an example process **1000** for user input detection by a case for an electronic device, in accordance with implementations of the subject technology. For explanatory purposes, the process **1000** is primarily described herein with reference to the case **112** of FIGS. **1-7**. However, the process **1000** is not limited to the case **112** of FIGS. **1-7**, and one or more blocks (or operations) of the process **1000** may be performed by one or more other components of other suitable devices, including the electronic device **110**, the electronic device **115**, and/or the servers **120**. Further for explanatory purposes, some of the blocks of the process **1000** are described herein as occurring in serial, or linearly. However, multiple blocks of the process **1000** may occur in parallel. In addition, the blocks of the process **1000** need not be performed in the order shown and/or one or more blocks of the process **1000** need not be

performed and/or can be replaced by other operations.

[0075] As illustrated in FIG. **10**, at block **1002**, a case (e.g., case **112**) having an internal cavity (e.g., cavity **270**) for storing for an electronic device (e.g., electronic device **204**, such as a media output device **150**, such as an earbud) may determine (**700**) that the electronic device is not present in the case.

[0076] At block **1004**, the case may obtain, responsive to determining that the electronic device is not present in the case, sensor data (e.g., sensor data **702**) from a sensor (e.g., sensor **220**, such as an accelerometer) of the case. In one or more implementations, the case may be free of sensors other than the sensor, and free of any sensors or input components on an exterior surface (e.g., outer surface **299**) of the case. In one or more implementations, obtaining the sensor data may include obtaining the sensor data responsive to an instruction (e.g., instruction **402**) from the other electronic device.

[0077] At block **1006**, the case (e.g., processor **224**) may provide the sensor data as an input to a machine learning model (e.g., machine learning model **704**) at the case. For example, the machine learning model may be executed by one or more processors (e.g., processor **224**) of the case, and may have been trained to detect one or more user inputs based on sensor data corresponding to the sensor of the case.

[0078] At block **1008**, the case (e.g., processor **224**) may identify (**703**), based on an output from the machine learning model, a user input on an exterior surface (e.g., outer surface **299**) of the case. For example, the output of the machine learning model may be a binary signal indicating detection or no detection of the user input (e.g., any user input or a particular user input for which the machine learning model has been trained), or may be a more complex output, such as a classification of one or more user inputs (e.g., as taps, multi-taps, touches, shakes, swipes, etc.) and/or a confidence or probability of the detection of the one or more user inputs.

[0079] In one or more implementations, a case that is free of sensors or input components on an exterior surface thereof may include, on the exterior surface, one or more features, such as a contrastive texture or material, that may provide a haptic feedback (e.g., indicating a particular location on the case at which a user can tap the case to provide an input as described herein). For example, the one or more features may be provided at a location that has been determined to enhance the effect of a tap on the sensor of the case.

[0080] At block **1010**, the case may perform a predetermined action (e.g., action **710**) responsive to the identifying of the user input on the exterior surface of the case. For example, the predetermined action may include a factory reset operation for the case. As another example, the predetermined action may include a locating operation for locating the electronic device. For example, the locating operation may include transmitting a signal **712** (e.g., a wireless signal) to the electronic device and/or receiving a signal **714** (e.g., a wireless signal) from the electronic device, and determining a location of the electronic device relative to the case based on the signal **712** and/or the signal **714**.

[0081] Various examples are described herein in which the case **112** is free of any user input components (e.g., buttons, switches, touch sensors, etc.) on the outer surface of the case, and one or more accelerometers and/or one or more microphones of the media output device(s) **150** that are stored in the case (e.g., and/or one or more accelerometers of the case **112** itself) can be used to detect user inputs to the case. However, it is also appreciated that one or more accelerometers and/or one or more microphones of the media output device(s) **150** that are stored in the case (e.g., and/or one or more accelerometers of the case **112** itself) can be used to detect user inputs to the case even in implementations in which the outer surface of the case **112** does include user input components, such as touch sensors, buttons, or switches. In these other implementations, the determination of whether a touch input occurred may be performed independent of any input from the touch sensors, buttons, or switches (e.g., in a use case in which the touch sensors, buttons, or switch on the case are deactivated or disabled, or in a use case in which the case includes a touch sensor, button, and/or switch and the microphone(s) and/or accelerometer(s) provide functionality

of another touch sensor, button, and/or switch that is not physically present on the case).

[0082] As described above, one aspect of the present technology is the gathering and use of data available from specific and legitimate sources for remote surface detection for electronic devices. The present disclosure contemplates that in some instances, this gathered data may include personal information data that uniquely identifies or can be used to identify a specific person. Such personal information data can include audio data, voice samples, voice profiles, demographic data, location-based data, online identifiers, telephone numbers, email addresses, home addresses, biometric data or records relating to a user's health or level of fitness (e.g., vital signs measurements, medication information, exercise information), date of birth, or any other personal information.

[0083] The present disclosure recognizes that the use of such personal information data, in the present technology, can be used to the benefit of users. For example, the personal information data can be used for remote surface detection for electronic devices.

[0084] The present disclosure contemplates that those entities responsible for the collection, analysis, disclosure, transfer, storage, or other use of such personal information data will comply with well-established privacy policies and/or privacy practices. In particular, such entities would be expected to implement and consistently apply privacy practices that are generally recognized as meeting or exceeding industry or governmental requirements for maintaining the privacy of users. Such information regarding the use of personal data should be prominently and easily accessible by users, and should be updated as the collection and/or use of data changes. Personal information from users should be collected for legitimate uses only. Further, such collection/sharing should occur only after receiving the consent of the users or other legitimate basis specified in applicable law. Additionally, such entities should consider taking any needed steps for safeguarding and securing access to such personal information data and ensuring that others with access to the personal information data adhere to their privacy policies and procedures. Further, such entities can subject themselves to evaluation by third parties to certify their adherence to widely accepted privacy policies and practices. In addition, policies and practices should be adapted for the particular types of personal information data being collected and/or accessed and adapted to applicable laws and standards, including jurisdiction-specific considerations which may serve to impose a higher standard. For instance, in the US, collection of or access to certain health data may be governed by federal and/or state laws, such as the Health Insurance Portability and Accountability Act (HIPAA); whereas health data in other countries may be subject to other regulations and policies and should be handled accordingly.

[0085] Despite the foregoing, the present disclosure also contemplates embodiments in which users selectively block the use of, or access to, personal information data. That is, the present disclosure contemplates that hardware and/or software elements can be provided to prevent or block access to such personal information data. For example, in the example of remote surface detection for electronic devices, the present technology can be configured to allow users to select to “opt in” or “opt out” of participation in the collection and/or sharing of personal information data during registration for services or anytime thereafter. In addition to providing “opt in” and “opt out” options, the present disclosure contemplates providing notifications relating to the access or use of personal information. For instance, a user may be notified upon downloading an app that their personal information data will be accessed and then reminded again just before personal information data is accessed by the app.

[0086] Moreover, it is the intent of the present disclosure that personal information data should be managed and handled in a way to minimize risks of unintentional or unauthorized access or use. Risk can be minimized by limiting the collection of data and deleting data once it is no longer needed. In addition, and when applicable, including in certain health related applications, data de-identification can be used to protect a user's privacy. De-identification may be facilitated, when appropriate, by removing identifiers, controlling the amount or specificity of data stored (e.g., collecting location data at city level rather than at an address level or at a scale that is insufficient

for facial recognition), controlling how data is stored (e.g., aggregating data across users), and/or other methods such as differential privacy.

[0087] Therefore, although the present disclosure broadly covers use of personal information data to implement one or more various disclosed embodiments, the present disclosure also contemplates that the various embodiments can also be implemented without the need for accessing such personal information data. That is, the various embodiments of the present technology are not rendered inoperable due to the lack of all or a portion of such personal information data.

[0088] FIG. 11 illustrates an electronic system **1100** with which one or more implementations of the subject technology may be implemented. The electronic system **1100** can be, and/or can be a part of, the media output device **150**, the electronic device **104**, the electronic device **110**, the electronic device **115**, and/or the server **120** as shown in FIG. 1. The electronic system **1100** may include various types of computer readable media and interfaces for various other types of computer readable media. The electronic system **1100** includes a bus **1108**, one or more processing unit(s) **1112**, a system memory **1104** (and/or buffer), a ROM **1110**, a permanent storage device **1102**, an input device interface **1114**, an output device interface **1106**, and one or more network interfaces **1116**, or subsets and variations thereof.

[0089] The bus **1108** collectively represents all system, peripheral, and chipset buses that communicatively connect the numerous internal devices of the electronic system **1100**. In one or more implementations, the bus **1108** communicatively connects the one or more processing unit(s) **1112** with the ROM **1110**, the system memory **1104**, and the permanent storage device **1102**. From these various memory units, the one or more processing unit(s) **1112** retrieves instructions to execute and data to process in order to execute the processes of the subject disclosure. The one or more processing unit(s) **1112** can be a single processor or a multi-core processor in different implementations.

[0090] The ROM **1110** stores static data and instructions that are needed by the one or more processing unit(s) **1112** and other modules of the electronic system **1100**. The permanent storage device **1102**, on the other hand, may be a read-and-write memory device. The permanent storage device **1102** may be a non-volatile memory unit that stores instructions and data even when the electronic system **1100** is off. In one or more implementations, a mass-storage device (such as a magnetic or optical disk and its corresponding disk drive) may be used as the permanent storage device **1102**.

[0091] In one or more implementations, a removable storage device (such as a floppy disk, flash drive, and its corresponding disk drive) may be used as the permanent storage device **1102**. Like the permanent storage device **1102**, the system memory **1104** may be a read-and-write memory device. However, unlike the permanent storage device **1102**, the system memory **1104** may be a volatile read-and-write memory, such as random access memory. The system memory **1104** may store any of the instructions and data that one or more processing unit(s) **1112** may need at runtime. In one or more implementations, the processes of the subject disclosure are stored in the system memory **1104**, the permanent storage device **1102**, and/or the ROM **1110** (which are each implemented as a non-transitory computer-readable medium). From these various memory units, the one or more processing unit(s) **1112** retrieves instructions to execute and data to process in order to execute the processes of one or more implementations.

[0092] The bus **1108** also connects to the input and output device interfaces **1114** and **1106**. The input device interface **1114** enables a user to communicate information and select commands to the electronic system **1100**. Input devices that may be used with the input device interface **1114** may include, for example, alphanumeric keyboards and pointing devices (also called “cursor control devices”). The output device interface **1106** may enable, for example, the display of images generated by electronic system **1100**. Output devices that may be used with the output device interface **1106** may include, for example, printers and display devices, such as a liquid crystal display (LCD), a light emitting diode (LED) display, an organic light emitting diode (OLED)

display, a flexible display, a flat panel display, a solid state display, a projector, or any other device for outputting information. One or more implementations may include devices that function as both input and output devices, such as a touchscreen. In these implementations, feedback provided to the user can be any form of sensory feedback, such as visual feedback, auditory feedback, or tactile feedback; and input from the user can be received in any form, including acoustic, speech, or tactile input.

[0093] Finally, as shown in FIG. 11, the bus 1108 also couples the electronic system 1100 to one or more networks and/or to one or more network nodes, such as the electronic device 110 shown in FIG. 1, through the one or more network interface(s) 1116. In this manner, the electronic system 1100 can be a part of a network of computers (such as a LAN, a wide area network (“WAN”), or an Intranet, or a network of networks, such as the Internet. Any or all components of the electronic system 1100 can be used in conjunction with the subject disclosure.

[0094] These functions described above can be implemented in computer software, firmware or hardware. The techniques can be implemented using one or more computer program products. Programmable processors and computers can be included in or packaged as mobile devices. The processes and logic flows can be performed by one or more programmable processors and by one or more programmable logic circuitry. General and special purpose computing devices and storage devices can be interconnected through communication networks.

[0095] Some implementations include electronic components, such as microprocessors, storage and memory that store computer program instructions in a machine-readable or computer-readable medium (also referred to as computer-readable storage media, machine-readable media, or machine-readable storage media). Some examples of such computer-readable media include RAM, ROM, read-only compact discs (CD-ROM), recordable compact discs (CD-R), rewritable compact discs (CD-RW), read-only digital versatile discs (e.g., DVD-ROM, dual-layer DVD-ROM), a variety of recordable/rewritable DVDs (e.g., DVD-RAM, DVD-RW, DVD+RW, etc.), flash memory (e.g., SD cards, mini-SD cards, micro-SD cards, etc.), magnetic and/or solid state hard drives, read-only and recordable Blu-Ray® discs, ultra density optical discs, any other optical or magnetic media, and floppy disks. The computer-readable media can store a computer program that is executable by at least one processing unit and includes sets of instructions for performing various operations. Examples of computer programs or computer code include machine code, such as is produced by a compiler, and files including higher-level code that are executed by a computer, an electronic component, or a microprocessor using an interpreter.

[0096] While the above discussion primarily refers to microprocessor or multi-core processors that execute software, some implementations are performed by one or more integrated circuits, such as application specific integrated circuits (ASICs) or field programmable gate arrays (FPGAs). In some implementations, such integrated circuits execute instructions that are stored on the circuit itself.

[0097] As used in this specification and any claims of this application, the terms “computer”, “server”, “processor”, and “memory” all refer to electronic or other technological devices. These terms exclude people or groups of people. For the purposes of the specification, the terms display or displaying means displaying on an electronic device. As used in this specification and any claims of this application, the terms “computer readable medium” and “computer readable media” are entirely restricted to tangible, physical objects that store information in a form that is readable by a computer. These terms exclude any wireless signals, wired download signals, and any other ephemeral signals.

[0098] To provide for interaction with a user, implementations of the subject matter described in this specification can be implemented on a computer having a display device, e.g., a CRT (cathode ray tube) or LCD (liquid crystal display) monitor, for displaying information to the user and a keyboard and a pointing device, e.g., a mouse or a trackball, by which the user can provide input to the computer. Other kinds of devices can be used to provide for interaction with a user as well; e.g.,

feedback provided to the user can be any form of sensory feedback, e.g., visual feedback, auditory feedback, or tactile feedback; and input from the user can be received in any form, including acoustic, speech, or tactile input. In addition, a computer can interact with a user by sending documents to and receiving documents from a device that is used by the user; e.g., by sending web pages to a web browser on a user's client device in response to requests received from the web browser.

[0099] Embodiments of the subject matter described in this specification can be implemented in a computing system that includes a back end component, e.g., as a data server, or that includes a middleware component, e.g., an application server, or that includes a front end component, e.g., a client computer having a graphical user interface or a Web browser through which a user can interact with an implementation of the subject matter described in this specification, or any combination of one or more such back end, middleware, or front end components. The components of the system can be interconnected by any form or medium of digital data communication, e.g., a communication network. Examples of communication networks include a local area network (“LAN”) and a wide area network (“WAN”), an inter-network (e.g., the Internet), and peer-to-peer networks (e.g., ad hoc peer-to-peer networks).

[0100] The computing system can include clients and servers. A client and server are generally remote from each other and may interact through a communication network. The relationship of client and server arises by virtue of computer programs running on the respective computers and having a client-server relationship to each other. In some embodiments, a server transmits data (e.g., an HTML page) to a client device (e.g., for purposes of displaying data to and receiving user input from a user interacting with the client device). Data generated at the client device (e.g., a result of the user interaction) can be received from the client device at the server.

[0101] Those of skill in the art would appreciate that the various illustrative blocks, modules, elements, components, methods, and algorithms described herein may be implemented as electronic hardware, computer software, or combinations of both. To illustrate this interchangeability of hardware and software, various illustrative blocks, modules, elements, components, methods, and algorithms have been described above generally in terms of their functionality. Whether such functionality is implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system. The described functionality may be implemented in varying ways for each particular application. Various components and blocks may be arranged differently (e.g., arranged in a different order, or partitioned in a different way) all without departing from the scope of the subject technology.

[0102] It is understood that the specific order or hierarchy of steps in the processes disclosed is an illustration of example approaches. Based upon design preferences, it is understood that the specific order or hierarchy of steps in the processes may be rearranged. Some of the steps may be performed simultaneously. The accompanying method claims present elements of the various steps in a sample order, and are not meant to be limited to the specific order or hierarchy presented.

[0103] The previous description is provided to enable any person skilled in the art to practice the various aspects described herein. The previous description provides various examples of the subject technology, and the subject technology is not limited to these examples. Various modifications to these aspects will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other aspects. Thus, the claims are not intended to be limited to the aspects shown herein, but is to be accorded the full scope consistent with the language claims, wherein reference to an element in the singular is not intended to mean “one and only one” unless specifically so stated, but rather “one or more.” Unless specifically stated otherwise, the term “some” refers to one or more. Pronouns in the masculine (e.g., his) include the feminine and neuter gender (e.g., her and its) and vice versa. Headings and subheadings, if any, are used for convenience only and do not limit the invention described herein.

[0104] The predicate words “configured to”, “operable to”, and “programmed to” do not imply any

particular tangible or intangible modification of a subject, but, rather, are intended to be used interchangeably. For example, a processor configured to monitor and control an operation or a component may also mean the processor being programmed to monitor and control the operation or the processor being operable to monitor and control the operation. Likewise, a processor configured to execute code can be construed as a processor programmed to execute code or operable to execute code.

[0105] The term automatic, as used herein, may include performance by a computer or machine without user intervention; for example, by instructions responsive to a predicate action by the computer or machine or other initiation mechanism. The word “example” is used herein to mean “serving as an example or illustration.” Any aspect or design described herein as “example” is not necessarily to be construed as preferred or advantageous over other aspects or designs.

[0106] A phrase such as an “aspect” does not imply that such aspect is essential to the subject technology or that such aspect applies to all configurations of the subject technology. A disclosure relating to an aspect may apply to all configurations, or one or more configurations. An aspect may provide one or more examples. A phrase such as an aspect may refer to one or more aspects and vice versa. A phrase such as an “embodiment” does not imply that such embodiment is essential to the subject technology or that such embodiment applies to all configurations of the subject technology. A disclosure relating to an embodiment may apply to all embodiments, or one or more embodiments. An embodiment may provide one or more examples. A phrase such as an “embodiment” may refer to one or more embodiments and vice versa. A phrase such as a “configuration” does not imply that such configuration is essential to the subject technology or that such configuration applies to all configurations of the subject technology. A disclosure relating to a configuration may apply to all configurations, or one or more configurations. A configuration may provide one or more examples. A phrase such as a “configuration” may refer to one or more configurations and vice versa.

[0107] All structural and functional equivalents to the elements of the various aspects described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and are intended to be encompassed by the claims. Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the claims. No claim element is to be construed under the provisions of 35 U.S.C. § 112 (f), unless the element is expressly recited using the phrase “means for” or, in the case of a method claim, the element is recited using the phrase “step for”.

Claims

1. A method, comprising: determining, by a case having an internal cavity for storing for an electronic device, that the electronic device is present in the case; providing, by the case, a first instruction to the electronic device to perform one or more detection operations for detecting a user input to the case; receiving, by the case from the electronic device, an indication of the user input to the case; and providing, by the case, a second instruction to the electronic device to perform an action responsive to the detection of the user input.
2. The method of claim 1, wherein an outer surface of the case is free of user input components.
3. The method of claim 2, wherein the user input to the case is a touch input on the outer surface of the case.
4. The method of claim 3, wherein the touch input comprises one or more taps on the outer surface of the case.
5. The method of claim 1, wherein the electronic device comprises an audio output device, and wherein the case is configured to charge a battery of the audio output device when the audio output device is in the case.
6. The method of claim 1, wherein the case comprises an accelerometer, and wherein the method

further comprises providing, by the case, accelerometer data from the accelerometer to the electronic device for use in the one or more detection operations, wherein the indication of the user input to the case is based on the accelerometer data and additional accelerometer data obtained by an additional accelerometer of the electronic device.

7. The method of claim 1, wherein the second instruction comprises an instruction, to the electronic device from the case, to start a pairing operation after a predetermined amount of time.

8. The method of claim 7, further comprising: receiving, by the case from the electronic device, an additional indication of an additional user input to the case prior to passage of the predetermined amount of time; and instructing the electronic device to perform an updated action, different from the action, if a further additional user input to the case is detected by the electronic device within an updated predetermined amount of time.

9. The method of claim 8, wherein the updated action comprises a factory reset of at least one of the case or the electronic device.

10. The method of claim 1, wherein the determining is performed responsive to a detection, by the case, of an opening of a lid of the case, and wherein the case is configured to provide a visual, audio, or haptic feedback responsive to the indication of the user input.

11. The method of claim 1, wherein the determining, by the case, comprises determining that the electronic device and an other electronic device are present in the case, and wherein the method further comprises determining, by the case and prior to providing the first instruction to the electronic device, which of the electronic device and the other electronic device is to perform the one or more detection operations.

12. The method of claim 11, further comprising providing, by the case to the other electronic device, the first instruction to perform the one or more detection operations in parallel with the one or more detection operations at the electronic device.

13. A method, comprising: obtaining sensor data from one or more sensors of an electronic device that is disposed within, and removable from, an other electronic device; and determining, by processing circuitry of the electronic device, whether the sensor data indicates a user input on an outer surface of the other electronic device.

14. The method of claim 13, further comprising providing, from the electronic device to the other electronic device within which the electronic device is disposed, an indication of the user input.

15. The method of claim 13, wherein obtaining the sensor data from the one or more sensors comprises obtaining the sensor data from at least an accelerometer and a microphone.

16. The method of claim 15, wherein determining whether the sensor data indicates the user input on the outer surface of the other electronic device comprises: providing the sensor data to a machine learning model at the electronic device; and determining whether the sensor data indicates the user input on the outer surface of the other electronic device based on an output of the machine learning model.

17. The method of claim 13, further comprising: receiving, by the electronic device from an additional electronic device disposed within and removable from the other electronic device, additional sensor data; and determining whether the sensor data indicates the user input on the outer surface of the other electronic device based on the sensor data and the additional sensor data.

18. The method of claim 13, wherein obtaining the sensor data comprises obtaining the sensor data responsive to an instruction from the other electronic device, wherein the electronic device comprises an audio output device, and wherein the other electronic device comprises a case for the electronic device.

19. A method, comprising: determining, by a case having an internal cavity for storing for an electronic device, that the electronic device is not present in the case; obtaining, responsive to determining that the electronic device is not present in the case, sensor data from a sensor of the case; providing the sensor data as an input to a machine learning model at the case; identifying, based on an output from the machine learning model, a user input on an exterior surface of the

case; and performing, by the case, a predetermined action responsive to the identifying of the user input on the exterior surface of the case.

20. The method of claim 19, wherein the predetermined action comprises a factory reset operation for the case.

21. The method of claim 19, wherein the predetermined action comprises a locating operation for locating the electronic device.
